

Decision Table Explanation

This document explains five decision tables covering common application workflows. Each table lists the governing conditions and rules, the resulting actions, and the test cases derived from every rule.

1. Library Book Reservation

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CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Book Available	Y	Y	N	Y	Y
C2: User Registered	Y	N	Y	Y	Y
C3: No Pending Dues	Y	-	-	N	Y
C4: Reservation Period Open	Y	-	-	-	N
C5: Membership Active	Y	-	-	-	-
ACTIONS					
A1: Reservation Confirmed	✓	✗	✗	✗	✗
A2: Show "Registration Required"	✗	✓	✗	✗	✗
A3: Show "Book Unavailable"	✗	✗	✓	✗	✗
A4: Show "Dues Pending"	✗	✗	✗	✓	✗
A5: Show "Reservation Closed"	✗	✗	✗	✗	✓

Note: Y = Yes, N = No, - = Not applicable, ✓ = Action is executed, ✗ = Action is not executed

Decision Table Conditions

- C1: Book Available
- C2: User Registered
- C3: No Pending Dues
- C4: Reservation Period Open
- C5: Membership Active

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC1_01	Book available, user registered, no dues, reservation period open, membership active	Reservation confirmed	R1
TC1_02	Book available, user not registered	Show 'Registration Required'	R2
TC1_03	Book not available, user registered	Show 'Book Unavailable'	R3
TC1_04	Registered user with pending dues	Show 'Dues Pending'	R4
TC1_05	Reservation period closed	Show 'Reservation Closed'	R5

Individual Explanation of Each Test Case

- TC1_01: Verifies that reservation confirmed when the conditions defined in R1 are satisfied.
- TC1_02: Verifies that show 'registration required' when the conditions defined in R2 are satisfied.
- TC1_03: Verifies that show 'book unavailable' when the conditions defined in R3 are satisfied.
- TC1_04: Verifies that show 'dues pending' when the conditions defined in R4 are satisfied.
- TC1_05: Verifies that show 'reservation closed' when the conditions defined in R5 are satisfied.

2. Flight Ticket Booking

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CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Seats Available	Y	N	Y	Y	Y
C2: Passport Valid	Y	-	N	Y	Y
C3: Payment Successful	Y	-	-	N	Y
C4: Travel Insurance Added	Y	-	-	-	-
C5: Check-in Window Open	Y	-	-	-	N
ACTIONS					
A1: Ticket Booked	✓	✗	✗	✗	✗
A2: Show "No Seats Available"	✗	✓	✗	✗	✗
A3: Show "Invalid Passport"	✗	✗	✓	✗	✗
A4: Show "Payment Failed"	✗	✗	✗	✓	✗
A5: Show "Check-in Closed"	✗	✗	✗	✗	✓

Note: Y = Yes, N = No, - = Not applicable, ✓ = Action is executed, ✗ = Action is not executed

Decision Table Conditions

- C1: Seats Available
- C2: Passport Valid
- C3: Payment Successful
- C4: Travel Insurance Added
- C5: Check-in Window Open

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC2_01	All conditions satisfied	Ticket booked	R1
TC2_02	No seats available	Show 'No Seats Available'	R2
TC2_03	Invalid passport	Show 'Invalid Passport'	R3
TC2_04	Payment failed	Show 'Payment Failed'	R4
TC2_05	Check-in window closed	Show 'Check-in Closed'	R5

Individual Explanation of Each Test Case

- TC2_01: Verifies that ticket booked when the conditions defined in R1 are satisfied.
- TC2_02: Verifies that show 'no seats available' when the conditions defined in R2 are satisfied.
- TC2_03: Verifies that show 'invalid passport' when the conditions defined in R3 are satisfied.
- TC2_04: Verifies that show 'payment failed' when the conditions defined in R4 are satisfied.

- TC2_05: Verifies that show 'check-in closed' when the conditions defined in R5 are satisfied.

3. Gym Membership Signup

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CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Slot Available	Y	N	Y	Y	Y
C2: Age Requirement Met	Y	-	N	Y	Y
C3: Payment Successful	Y	-	-	N	Y
C4: Health Declaration Submitted	Y	-	-	-	N
C5: Membership Plan Selected	Y	-	-	-	-
ACTIONS					
A1: Membership Activated	✓	✗	✗	✗	✗
A2: Show "Slot Full"	✗	✓	✗	✗	✗
A3: Show "Age Restriction"	✗	✗	✓	✗	✗
A4: Show "Payment Failed"	✗	✗	✗	✓	✗
A5: Show "Health Form Required"	✗	✗	✗	✗	✓

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Decision Table Conditions

- C1: Slot Available
- C2: Age Requirement Met
- C3: Payment Successful
- C4: Health Declaration Submitted
- C5: Membership Plan Selected

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC3_01	All conditions satisfied	Membership activated	R1
TC3_02	No slots available	Show 'Slot Full'	R2
TC3_03	Age requirement not met	Show 'Age Restriction'	R3
TC3_04	Payment failed	Show 'Payment Failed'	R4
TC3_05	Health declaration missing	Show 'Health Form Required'	R5

Individual Explanation of Each Test Case

- TC3_01: Verifies that membership activated when the conditions defined in R1 are satisfied.
- TC3_02: Verifies that show 'slot full' when the conditions defined in R2 are satisfied.
- TC3_03: Verifies that show 'age restriction' when the conditions defined in R3 are satisfied.
- TC3_04: Verifies that show 'payment failed' when the conditions defined in R4 are satisfied.

- TC3_05: Verifies that show 'health form required' when the conditions defined in R5 are satisfied.

4. ATM Cash Withdrawal

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CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Sufficient Balance	Y	N	Y	Y	Y
C2: Card Valid	Y	-	N	Y	Y
C3: PIN Correct	Y	-	-	N	Y
C4: Daily Limit Not Exceeded	Y	-	-	-	N
C5: ATM Has Cash	Y	-	-	-	-
ACTIONS					
A1: Cash Dispensed	✓	✗	✗	✗	✗
A2: Show "Insufficient Balance"	✗	✓	✗	✗	✗
A3: Show "Invalid Card"	✗	✗	✓	✗	✗
A4: Show "Incorrect PIN"	✗	✗	✗	✓	✗
A5: Show "Limit Exceeded"	✗	✗	✗	✗	✓

Note: Y = Yes, N = No, - = Not applicable, ✓ = Action is executed, ✗ = Action is not executed

Decision Table Conditions

- C1: Sufficient Balance
- C2: Card Valid
- C3: PIN Correct
- C4: Daily Limit Not Exceeded
- C5: ATM Has Cash

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC4_01	All conditions satisfied	Cash dispensed	R1
TC4_02	Insufficient balance	Show 'Insufficient Balance'	R2
TC4_03	Invalid card	Show 'Invalid Card'	R3
TC4_04	Incorrect PIN entered	Show 'Incorrect PIN'	R4
TC4_05	Daily limit exceeded	Show 'Limit Exceeded'	R5

Individual Explanation of Each Test Case

- TC4_01: Verifies that cash dispensed when the conditions defined in R1 are satisfied.
- TC4_02: Verifies that show 'insufficient balance' when the conditions defined in R2 are satisfied.
- TC4_03: Verifies that show 'invalid card' when the conditions defined in R3 are satisfied.
- TC4_04: Verifies that show 'incorrect PIN' when the conditions defined in R4 are satisfied.

- TC4_05: Verifies that show 'limit exceeded' when the conditions defined in R5 are satisfied.

5. Online Exam Submission

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CONDITIONS	RULES				
	R1	R2	R3	R4	R5
C1: Exam Window Open	Y	N	Y	Y	Y
C2: User Logged In	Y	-	N	Y	Y
C3: All Questions Answered	Y	-	-	N	Y
C4: Time Not Expired	Y	-	-	-	N
C5: Internet Connection Stable	Y	-	-	-	-
ACTIONS					
A1: Submission Successful	✓	✗	✗	✗	✗
A2: Show "Exam Closed"	✗	✓	✗	✗	✗
A3: Show "Login Required"	✗	✗	✓	✗	✗
A4: Show "Incomplete Answers"	✗	✗	✗	✓	✗
A5: Show "Time Expired"	✗	✗	✗	✗	✓

Note: Y = Yes, N = No, - = Not applicable, ✓ = Action is executed, ✗ = Action is not executed

Decision Table Conditions

- C1: Exam Window Open
- C2: User Logged In
- C3: All Questions Answered
- C4: Time Not Expired
- C5: Internet Connection Stable

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule
TC5_01	All conditions satisfied	Submission successful	R1
TC5_02	Exam window closed	Show 'Exam Closed'	R2
TC5_03	User not logged in	Show 'Login Required'	R3
TC5_04	Some questions unanswered	Show 'Incomplete Answers'	R4
TC5_05	Time expired before submission	Show 'Time Expired'	R5

Individual Explanation of Each Test Case

- TC5_01: Verifies that submission successful when the conditions defined in R1 are satisfied.
- TC5_02: Verifies that show 'exam closed' when the conditions defined in R2 are satisfied.
- TC5_03: Verifies that show 'login required' when the conditions defined in R3 are satisfied.
- TC5_04: Verifies that show 'incomplete answers' when the conditions defined in R4 are satisfied.

- TC5_05: Verifies that show 'time expired' when the conditions defined in R5 are satisfied.